

- Autonomous vehicles continuously collect data about how their environment is changing, adjusting the drive based on weather conditions, engine status, and countless other variables.
- In the medical field, sensors generate time data streams for blood pressure tracking, weight tracking, cholesterol measurements, heart rate monitoring, etc.

There are numerous real-world scenarios where we collect time series data streams. These demand an efficient database for storing and retrieving time series data.

Time series data analysis

Time series data is a sequence of data points collected over time intervals, giving us the ability to track changes over time. Because data points in time series are collected at adjacent time periods, the observations can be correlated. This feature distinguishes time series data from traditional data. Time series data can be useful to help recognise patterns or a trend. Knowing the value of a specific parameter at the current time is quite different than the ability to observe its behaviour over a long time interval. Time series data allows us to measure and analyse change — what has changed in the past, what is changing in the present, and what changes may take place in the future.

Time series data can track changes over milliseconds, days, or even years. Table 1 outlines the typical questions that time series analysis can help to answer.

Typical steps in time series data analysis are:

- Collecting the data and cleaning it
- Visualising with respect to time vs key feature
- Observing the stationarity of the series
- Developing charts to understand the nature of the data
- Model building such as AR, MA, ARMA and ARIMA
- Extracting insights from the predictions

There are three components of time series analysis — trend, seasonality and residual analysis.

Trend: This indicates the direction in which the data is moving over a period of time.

Seasonality: Seasonality is about periodic behaviour — spikes or drops caused by different factors like:

- Naturally occurring events like weather fluctuations
- Business or administrative procedures like the start or end of a fiscal year
- Social and cultural behaviour like holidays or festivals
- Calendar events, like the number of Mondays per month or holidays that change every year

Residual analysis: These are the irregular fluctuations that cannot be predicted using trend or seasonality analysis.

An observed data stream could be additive (trend + seasonality + residual) or multiplicative (trend * seasonality * residual).

Once these components are identified, models are built to understand time series and check for anomalies, forecasting and correlations. For time series data modelling, AR, MA, ARMA and ARIMA algorithms are widely adopted. Many other advanced AI/ML algorithms are being proposed for better evaluation.

Time series databases

A time series database (TSDB) is a database optimised for time-stamped or time series data. Time series data is simply measurements or events that are tracked, monitored, down sampled, and aggregated over time. These could be server metrics, application performance monitoring, network data, sensor data, events, clicks, trades in a market, and many other types of analytics data.

Looking back 10 years, the amount of data that was once collected in 10 minutes for some very active systems is now generated every second. To process these high volumes, we need different tools and approaches.

Table 1

Category	Typical questions to be addressed
Prognostication	What are the short- and long-term trends for a measurement or group of measurements?
Introspection	How do several measurements correlate over a period of time?
Prediction	How do I build a machine learning model based on the temporal behaviour of many measurements correlated to externally known facts?
Introspection	Have similar patterns of measurements preceded similar events?
Diagnosis	What measurements might indicate the cause of some event, such as a system failure?
Forecasting	How many more servers will be needed for handling next quarter's workload?
Segmentation	How to divide a data stream into a sequence of discrete segments in order to reveal the underlying properties of its source?